

13. Build Deployment

13.1 Introduction

Build Deployment is the process of packaging, validating, and preparing an application or automation framework for execution in different environments. In modern software development, deployment activities are integrated with CI/CD pipelines to ensure faster and more reliable software delivery.

For this project, the Parabank Automation Framework was developed using Selenium WebDriver, Java, TestNG, and Maven. The framework was integrated with GitHub Actions to automatically execute test cases whenever source code changes were pushed to the repository.

The deployment process focused on source code management, build creation, test execution, report generation, and continuous integration validation.

13.2 Purpose of Build Deployment

The purpose of build deployment is to:

- Validate the automation framework.
 - Ensure successful project build generation.
 - Execute automated test cases.
 - Verify framework stability.
 - Support continuous integration.
 - Generate execution reports.
 - Provide deployment readiness confirmation.
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13.3 Build Deployment Objectives

The objectives of deployment are:

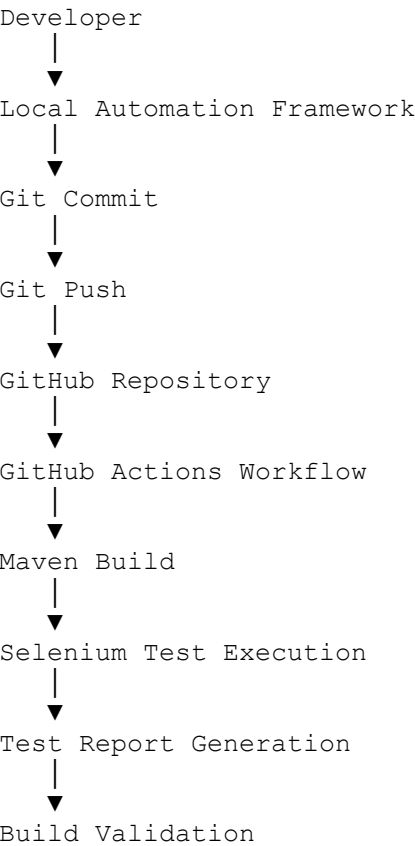
- Automate build generation.
 - Automate test execution.
 - Ensure successful integration of source code.
 - Detect defects early.
 - Generate automated reports.
 - Improve software quality.
 - Support continuous testing practices.
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13.4 Deployment Environment

The deployment activities were performed using the following environment:

Component	Details
Operating System	Windows 11
IDE	Eclipse IDE
Programming Language	Java 21
Automation Tool	Selenium WebDriver
Testing Framework	TestNG
Build Tool	Maven
Version Control	Git
Repository	GitHub
CI/CD Tool	GitHub Actions
Browser	Google Chrome

13.5 Deployment Architecture



13.6 Deployment Process

The deployment process followed the steps below.

Step 1: Framework Development

Automation scripts were developed using:

- Selenium WebDriver
- Java
- TestNG
- Page Object Model

Modules Automated:

- Login
 - Account Overview
 - Transfer Funds
 - Bill Payment
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Step 2: Local Build Validation

Before deployment, the framework was executed locally.

Command used:

```
mvn clean test
```

Activities performed:

- Maven build validation
- Dependency verification
- Test execution
- Report generation

Result:

Build executed successfully.

Step 3: Git Repository Initialization

Git repository was initialized.

Command:

```
git init
```

Purpose:

- Enable source control
 - Track project changes
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Step 4: Source Code Commit

Project files were committed to Git.

Command:

```
git add .  
git commit -m "Initial Commit - Parabank Automation Framework"
```

Purpose:

- Store project snapshot
 - Maintain version history
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Step 5: GitHub Repository Integration

Local repository was connected to GitHub.

Command:

```
git remote add origin <repository-url>
```

Purpose:

- Enable remote storage
 - Support CI/CD integration
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Step 6: Source Code Push

Code was pushed to GitHub.

Command:

```
git push origin main
```

Purpose:

- Upload source code

- Trigger CI/CD workflow
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Step 7: CI/CD Pipeline Execution

GitHub Actions automatically executed the workflow.

Activities:

- Repository checkout
 - Java setup
 - Maven build
 - Selenium execution
 - Report generation
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Step 8: Report Generation

Execution reports generated:

- TestNG Reports
- Execution Logs
- Screenshots

Artifacts stored in GitHub Actions.

13.7 Build Validation

The following validations were performed:

Validation	Status
Source Code Upload	Pass
Maven Build	Pass
Dependency Resolution	Pass
Selenium Execution	Pass
TestNG Execution	Pass
GitHub Actions Execution	Pass
Report Generation	Pass

13.8 Deployment Deliverables

The following deliverables were produced:

Source Code

- Automation Framework
- Test Scripts
- Utility Classes

CI/CD Deliverables

- GitHub Actions Workflow
- Build Logs
- Pipeline Reports

Testing Deliverables

- Test Execution Reports
- Screenshots
- Defect Reports

13.9 Benefits of Deployment

The deployment process provided several benefits:

- Automated build validation
- Automated testing
- Faster feedback
- Reduced manual effort
- Improved code quality
- Continuous integration support
- Reliable execution process

13.10 Deployment Outcome

The deployment activities were completed successfully.

Achievements:

- Automation framework deployed.
- GitHub repository configured.
- CI/CD pipeline integrated.
- Test execution automated.
- Reports generated successfully.

13.11 Conclusion

The build deployment process successfully integrated the Parabank Automation Framework with GitHub Actions. Automated builds, test execution, and report generation improved the overall quality assurance process. The deployment approach ensured reliable and repeatable execution while supporting continuous integration and automated testing practices.
